Agentic AI for Plant Disease Diagnosis: An Autonomous Multi-Agent Framework Integrating Vision, Retrieval, and Reasoning

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***Abstract*—Automated plant disease detection using convolutional neural networks has achieved strong classification accuracy on benchmark datasets such as PlantVillage, yet a standalone classifier provides only a label and a confidence score: it offers no explanation, no differential reasoning over plausible alternatives, and no mechanism to defer uncertain cases to a human expert. This paper presents an agentic artificial intelligence framework that wraps a ResNet-18 vision classifier with four cooperating agents – a Confidence and Image-Quality Agent, a Retrieval Agent, a Reasoning Agent, and an Orchestrator – to convert disease classification into an autonomous, explainable, and safety-aware diagnostic advisory system. The Orchestrator conditions expensive retrieval and reasoning computation on prediction confidence and class ambiguity, invoking a full retrieval-augmented reasoning pipeline only for uncertain cases and escalating low-confidence, low-quality inputs directly to human review rather than acting on them. We evaluate the system using both conventional classification metrics and a set of agentic metrics – correct-action rate, harmful-action rate, and human-intervention rate – that quantify the safety-utility trade-off of autonomous deferral relative to a naive always-trust-the-classifier baseline. Evaluation is deliberately conducted on an out-of-distribution test set of 190 leaf images collected from sources outside the PlantVillage training corpus, spanning all 38 classes, so that the reported numbers reflect generalization rather than in-distribution recall. Under this realistic distribution shift the classifiers, which reach near-perfect accuracy (up to 100%) on in-distribution same-source images, generalize far more weakly – individual models reach 20.5–23.2% top-1 accuracy and a three-model federated ensemble (FedAvg, FedOpt, FedMA) reaches 24.2% – yet it is exactly here that the agentic layer proves its worth: for the ensemble it reduces the harmful-action rate from a naive always-act baseline of 75.8% to 14.2%, a roughly five-fold reduction, at a human-intervention rate of 63.2%, by recognizing its own uncertainty and deferring unreliable cases to a human. These results support agent-orchestrated pipelines as a practical bridge between imperfect vision models and trustworthy, deployable agricultural decision support.**

***Index Terms*—Agentic AI, Plant Disease Detection, Retrieval-Augmented Generation, Multi-Agent Systems, PlantVillage, ResNet-18, Precision Agriculture, Autonomous Decision-Making**

I. Introduction

Plant diseases remain among the largest threats to global food security, with crop losses attributable to pests and pathogens estimated to reduce yields by 20–40% annually across major staple crops. Early and accurate diagnosis is the single most effective lever for limiting this loss, since most fungicide and cultural interventions are far more effective when applied before a disease reaches an advanced stage. The introduction of large, labeled image datasets such as PlantVillage [1], together with deep convolutional architectures, has made it possible to train classifiers that recognize dozens of crop-disease categories from a single leaf photograph with accuracy exceeding 95% on held-out benchmark data [2].

Despite this accuracy, a bare classifier is a narrow, reactive tool in the sense described by the broader agentic AI literature [3], [4]: it consumes an image and returns a label and a softmax score, with no capacity to explain *why* a diagnosis was reached, no mechanism to weigh alternative diagnoses when the image is ambiguous, and – most critically for a system whose output may drive real pesticide use – no way to recognize its own uncertainty and defer to a human expert. A classifier that is wrong 5% of the time will, if deployed naively, confidently recommend the wrong treatment in exactly those 5% of cases, with no distinguishing signal to separate them from the 95% it gets right.

Agentic artificial intelligence – systems characterized by autonomy, goal-directed behavior, tool use, and the capacity to plan multi-step actions in response to their environment – has been proposed as a paradigm for exactly this class of problem across domains such as healthcare, energy management, and climate modeling [3], [5]–[8]. Reviews of agentic

AI applications have surveyed its use extensively in these areas [9]–[11], yet its application to agricultural disease diagnosis as an autonomous, safety-aware decision-support agent has received comparatively little attention. Existing plant-disease work overwhelmingly frames the problem as pure image classification, leaving the reasoning, retrieval, and deferral behavior that would make such a system trustworthy for real-world deployment largely unaddressed.

This paper closes that gap. We present an agentic architecture that reuses a trained plant-disease classifier as a perception module and wraps it with four cooperating agents that together transform a single-shot classification into an autonomous diagnostic workflow. The specific contributions of this paper are:

1) We propose and implement a five-component agentic architecture – Vision, Confidence/Quality, Retrieval, Reasoning, and Orchestrator – that upgrades a standard CNN plant-disease classifier into an autonomous diagnostic advisory agent.
2) We design an orchestration policy that conditions retrieval and reasoning computation on prediction confidence and class ambiguity, so that computationally expensive reasoning is invoked only where it is needed, while low-confidence or low-quality inputs are escalated to a human rather than acted upon.
3) We introduce a set of agentic evaluation metrics – correct-action rate, harmful-action rate, and human-intervention rate – that capture the safety-utility trade-off of an autonomous diagnostic system in a way that classification accuracy alone cannot.
4) We report an empirical evaluation of the complete pipeline under a realistic distribution shift – on an out-of-distribution set of web-sourced leaf images rather than held-out PlantVillage images – quantifying the reduction in harmful-action rate achieved relative to a naive classifier-only baseline, for both single-model and ensemble Vision Agents.

The remainder of this paper is organized as follows. Section II reviews the foundations of agentic AI and positions this work relative to prior plant-disease classification research. Section III describes the proposed multi-agent architecture. Section IV details the implementation. Section V defines the experimental protocol and metrics. Section VI reports results, and Section VII concludes with directions for future work.

II. Related Work

A. Foundations of Agentic AI

Agentic AI refers to systems designed with autonomy, goal-directed behavior, and the capacity to take initiative, moving beyond passive tools that only respond to prompts [3], [4]. Four capabilities are commonly identified as its foundation. *Reinforcement learning* provides a framework for sequential decision-making under reward and penalty, prioritizing experiential learning over supervised label-fitting [12]. *Goal-oriented architectures* give an agent explicit objectives against which it can plan and prioritize actions, integrating symbolic reasoning and planning with learned components [13], [14]. *Autonomous decision-making* is the capacity to evaluate options and act without continuous human supervision, underpinned by perception, reasoning, and action-selection mechanisms [15]. *Environment interaction* closes the loop: an agent perceives state, updates an internal model, and takes actions that alter its environment, enabling adaptation to conditions that were not seen during training [16].

A closely related concept is agentic memory, commonly decomposed into procedural, short-term, long-term, episodic, and semantic memory [3]. Procedural memory stores reusable action sequences; short-term memory holds task-local context; long-term and semantic memory persist facts and preferences across sessions; and episodic memory records specific past interactions to inform future behavior. As detailed in Section III, the present system deliberately implements only the semantic-memory component required for retrieval-augmented reasoning, treating persistent episodic memory and longitudinal disease tracking as a scoped future extension (Section VII) rather than a claim made in the current implementation.

TABLE I
Traditional AI versus Agentic AI

Aspect	**Traditional AI**	**Agentic AI (this work)**
Output	Label + confidence score	Diagnosis, grounded explanation, treatment, and a deferral decision
Initiative	Reactive; always emits a prediction	Proactive; may withhold action and request human review
Reasoning	Single forward pass	Multi-step: perceive, assess, retrieve, reason, act
Uncertainty handling	None; wrong predictions are indistinguishable from correct ones	Explicit confidence and ambiguity gating via the Orchestrator
Explanation	None	Retrieved-passage-grounded justification

B. Traditional versus Agentic AI

Table I summarizes the distinction between the traditional classification paradigm that dominates existing plant-disease literature and the agentic paradigm adopted in this work. The key shift is from a system that is reactive and provides a single output, to one that is proactive, reasons over multiple candidate outcomes, and can decline to act.

C. Deep Learning for Plant Disease Detection

The PlantVillage dataset [1] and the benchmark results of Mohanty et al. [2] established convolutional neural networks as the dominant approach for image-based plant disease diagnosis, with subsequent work exploring deeper backbones, transfer learning, and federated training [17] across distributed agricultural data sources. This body of work optimizes exclusively for

classification accuracy; none of it, to our knowledge, addresses the downstream question of *what an automated system should do* with a prediction once it is made – whether to act on it, explain it, or defer it – which is the problem this paper addresses.

D. Agentic AI in Agriculture

Surveys of agentic AI applications have covered healthcare [3], [14], finance, cybersecurity, and climate and energy management in considerable depth [6], [7], [9], documenting autonomous agents that monitor, plan, and act with limited human oversight in each of these domains. Agriculture is frequently listed as a target domain for agentic AI in these broader surveys, typically in connection with resource optimization, irrigation scheduling, or supply-chain management at the farm or regional level. However, at the level of an individual diagnostic interaction between a farmer, a leaf photograph, and a recommended treatment, the literature remains dominated by single-model image classifiers. The architecture proposed in this paper is, to our knowledge, among the first to apply the full agentic pattern – perception, uncertainty-aware reasoning, tool-augmented retrieval, and autonomous deferral – to this specific and consequential diagnostic setting, where an incorrect autonomous recommendation carries a direct economic and food-security cost.

E. Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) couples a language model with a non-parametric retrieval step over an external corpus, grounding generated text in retrieved evidence rather than relying solely on parametric knowledge [18]. This mechanism is well suited to diagnostic explanation, where the treatment and prevention advice given to a farmer should be traceable to an authoritative agricultural source rather than generated from the language model's unverified prior knowledge. Section III describes how this work employs RAG as the semantic-memory substrate for the Reasoning Agent.

III. Proposed System Architecture

The system is composed of five cooperating components: a Vision Agent, a Confidence and Image-Quality Agent, a Retrieval Agent, a Reasoning Agent, and an Orchestrator that governs control flow between them. Fig. 1 illustrates the architecture and the three execution paths the Orchestrator can select.

A. Vision Agent

The Vision Agent maps an input leaf image to a probability distribution over the PlantVillage disease taxonomy using one or more ResNet-18 convolutional networks [19]. The backbones are reused from our prior federated learning study on plant disease classification, in which they were trained across simulated clients under a non-IID partition of the training data. The system supports two configurations: a single ResNet-18, and an ensemble of three ResNet-18 checkpoints produced by distinct federated aggregation strategies – Federated

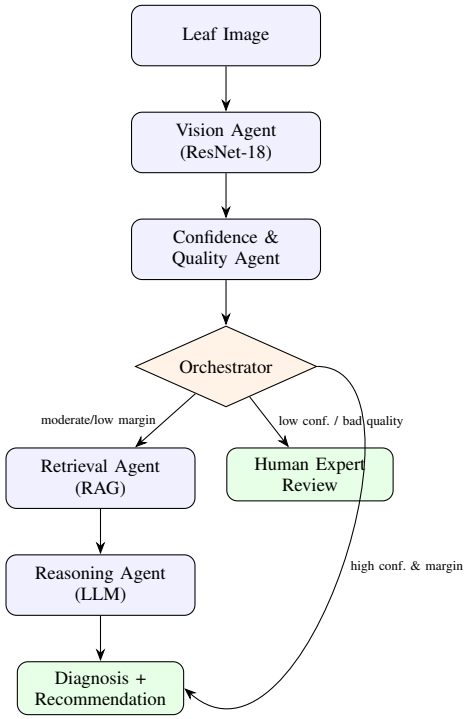


Fig. 1. Proposed agentic architecture. The Orchestrator conditionally routes each case to a fast direct-answer path, the full retrieval-and-reasoning path, or human escalation.

Averaging (FedAvg) [17], adaptive Federated Optimization (FedOpt) [20], and Federated Matched Averaging (FedMA) [21] – whose softmax distributions are averaged. The ensemble additionally yields an *agreement ratio*: the fraction of member models whose individual top-1 prediction matches the averaged ensemble label. This serves as a model-based uncertainty signal complementary to the softmax confidence, since independently trained models can diverge on a hard case even when the averaged confidence is high. In the present system the Vision Agent is used purely as a perception module: it emits the predicted label, the full class-probability distribution, and – in ensemble mode – the agreement ratio, all of which are passed to the next agent.

B. Confidence and Image-Quality Agent

This agent performs lightweight, non-parametric post-processing on the Vision Agent's output and on the input image itself, producing a structured reliability signal rather than a new prediction. Up to four quantities are computed: (i) *confidence*, the top-1 softmax probability; (ii) *margin*, the gap between the top-1 and top-2 class probabilities, which captures ambiguity between visually similar diseases that raw confidence alone can miss; (iii) a binary *quality flag*, derived from a Laplacian-variance blur check and a brightness-range check on the input image, to catch cases where the image itself – rather than the disease – is the source of uncertainty;

and (iv) when the Vision Agent runs as an ensemble, the *agreement ratio* propagated from it, which flags cases where independently trained models disagree even if the averaged confidence is high. These signals are passed to the Orchestrator as a single structured object.

C. Retrieval Agent

The Retrieval Agent implements retrieval-augmented generation over a static knowledge base curated from publicly available agricultural extension and plant-pathology reference material, covering the crops and diseases represented in the PlantVillage taxonomy. Each reference document is embedded and indexed for dense semantic retrieval. Given a predicted disease label (and, when the margin is low, its runner-up candidate), the Retrieval Agent returns the top-k most relevant passages, which serve as grounding evidence for the Reasoning Agent. This agent is invoked only when the Orchestrator determines that a case warrants full reasoning, rather than on every prediction.

D. Reasoning Agent

The Reasoning Agent is a large language model prompted with the predicted label, its confidence and margin, and the passages returned by the Retrieval Agent. It produces three outputs: (i) a diagnosis statement identifying the most likely disease and, when margin is low, up to two plausible alternatives; (ii) an explanation that is required to cite which retrieved passage supports each factual claim, so that the output can be checked for grounding rather than accepted as an unverifiable model assertion; and (iii) a treatment and prevention recommendation drawn from the retrieved evidence rather than generated from unverified parametric knowledge.

E. Orchestrator

The Orchestrator is the component that makes the system agentic rather than a fixed pipeline: it decides, per case, which of the other agents to invoke. Its policy is summarized in Table II. High-confidence, high-margin, good-quality cases bypass retrieval and reasoning entirely and are answered directly from a lookup, minimizing latency and inference cost. Moderate-confidence or low-margin cases trigger the full retrieval-and-reasoning pipeline. Low-confidence or low-quality cases are escalated directly to human review without any automated treatment recommendation being issued, since an automated system should not offer possibly harmful advice under conditions where it cannot support that advice. In ensemble mode, heavy disagreement among the member models (agreement ratio below a threshold) is treated as an additional escalation trigger, since divergence between independently trained models is itself evidence that the case is unreliable, even when the averaged softmax confidence is high.

Algorithm 1 formalizes this routing logic.

F. Scope: Memory and Longitudinal Tracking

An episodic Memory Agent that logs each diagnosis and a Follow-up Agent that requests a subsequent image to track disease progression over time are natural extensions of this architecture, consistent with the long-term and episodic memory categories discussed in Section II. We deliberately scope them out of the present implementation, both because they require longitudinal image data not available in the PlantVillage benchmark and to keep the system within a tractable, rigorously evaluable footprint; they are discussed further as future work in Section VII.

TABLE II
ORCHESTRATOR ROUTING POLICY

Condition	Action
High confidence, high margin, model agreement, good image quality	Direct lookup-based recommendation; retrieval and reasoning skipped
Moderate confidence, or low margin, good image quality	Full pipeline: retrieval $\rightarrow$ reasoning $\rightarrow$ grounded recommendation with caveats
Low confidence, poor image quality, or heavy model disagreement	Escalate to human expert; no automated recommendation issued

Algorithm 1 Orchestrator Routing Policy

Require: image x; thresholds $\tau_{high} = 0.85$, $\tau_{low} = 0.55$, $\tau_{margin} = 0.15$, $\tau_{agree} = 0.5$
$\hat{y}, p, a \leftarrow$ VisionAgent(x) {label, softmax vector, agreement ratio}
$c \leftarrow \max(p)$; $m \leftarrow p_{(1)} - p_{(2)}$ {confidence, margin}
$q \leftarrow$ QualityCheck(x) {blur / brightness check}
if $c \geq \tau_{high}$ **and** $m \geq \tau_{margin}$ **and** $a \geq \tau_{agree}$ **and** $q =$ good **then**
  return DirectLookup$(\hat{y})$ {skip retrieval & reasoning}
else if $c \geq \tau_{low}$ **and** $a \geq \tau_{agree}$ **and** $q =$ good **then**
  $D \leftarrow$ RetrievalAgent$(\hat{y})$
  return ReasoningAgent$(\hat{y}, c, m, D)$
else
  return EscalateToHuman$(x, \hat{y}, c, q, a)$
end if

IV. IMPLEMENTATION DETAILS

Dataset. We use the PlantVillage dataset [1], comprising leaf images spanning multiple crop species and disease categories, split into training, validation, and held-out test partitions consistent with our prior federated learning experiments on this dataset.

Vision backbone. ResNet-18 [19] is reused unmodified from prior work; no retraining was performed for this study, isolating the evaluation to the contribution of the agentic layer. We evaluate two Vision Agent configurations: a single FedAvg-trained checkpoint, and a three-model ensemble that combines checkpoints trained with FedAvg [17], FedOpt [20], and FedMA [21] aggregation by averaging their softmax outputs. All three checkpoints share the same 38-class ResNet-18 architecture and differ only in their federated aggregation strategy.

Orchestrator thresholds. The high-confidence, low-confidence, and margin thresholds are configured at 0.85, 0.55, and 0.15 respectively; in ensemble mode, an agreement-ratio threshold of 0.5 additionally escalates cases on which the member models substantially disagree. The thresholds are selected to bias the system toward caution: cases are escalated rather than acted upon whenever confidence, margin, or inter-model agreement falls outside a conservative operating range.

Retrieval index. Reference documents are chunked and embedded with the `bge-small-en-v1.5` sentence-embedding model and indexed in a FAISS vector store [22] for cosine-similarity retrieval; the Retrieval Agent returns the top-k ($k = 5$) passages per query.

Reasoning model. The Reasoning Agent is implemented with an instruction-tuned large language model accessed through an OpenAI-compatible API via a provider-agnostic wrapper, so that the model can be swapped without changing the rest of the system. It is prompted with a fixed template requiring passage-grounded citation for every factual claim in its output and instructed to decline to recommend a treatment that the retrieved evidence does not support.

Knowledge base. The static reference corpus underlying the Retrieval Agent comprises 125 curated documents drawn from open-access PubMed Central articles and agricultural extension / plant-pathology reference material, covering symptoms, causal pathogens, and management practices for the crop-disease categories present in the PlantVillage taxonomy. Chunking and embedding each document independently yields 16,706 indexed passage chunks, allowing the Retrieval Agent to return passage-level rather than document-level evidence to the Reasoning Agent.

Episodic logging. Every case processed by the Orchestrator, together with its routing decision and downstream output, is logged in a structured record. This log is used for offline evaluation (Section V) and, as discussed in Section VII, forms the basis on which a full episodic Memory Agent could be built in future work.

V. Experimental Setup

We evaluate the system on an out-of-distribution test set of 190 leaf images collected from public sources outside the PlantVillage dataset (five per class across all 38 crop-disease categories). Because the Vision Agent backbones were trained on PlantVillage, this set deliberately measures generalization under a realistic distribution shift – heterogeneous, web-sourced photographs rather than the controlled, uniform-background images the models were trained on – instead of in-distribution recall. We report results for each individual model (FedAvg, FedOpt, FedMA) and for the three-model ensemble, using two families of metrics.

Classification metrics. Top-1 accuracy and macro-averaged F1 score of the Vision Agent in isolation, to establish the baseline perceptual performance the agentic layer builds upon.

Agentic metrics. For each test case, let the system's action be either *act* (issue a treatment recommendation, whether via the direct-lookup or full-reasoning path) or *escalate* (defer to a human). We define:

$$\text{Correct-action rate} = \frac{|\{\,\text{act} \wedge \hat{y} = y\,\}|}{N} \tag{1}$$

$$\text{Harmful-action rate} = \frac{|\{\,\text{act} \wedge \hat{y} \neq y\,\}|}{N} \tag{2}$$

$$\text{Human-intervention rate} = \frac{|\{\,\text{escalate}\,\}|}{N} \tag{3}$$

where $\hat{y}$ is the predicted label, y is the ground-truth label, and N is the number of test cases. We additionally report the *naive baseline harmful-action rate*, defined as the classifier's raw error rate under a policy that always acts regardless of confidence – equivalent to deploying the Vision Agent alone without any orchestration. The reduction from this baseline to the agentic harmful-action rate is the primary indicator of the value added by the Confidence/Quality Agent and Orchestrator.

Grounding check. On the reasoning outputs from cases routed through the full pipeline, we qualitatively inspect whether cited factual claims are supported by the retrieved passages they reference and whether the agent declines to recommend treatments that the evidence does not contain. A formal, large-scale grounding annotation with inter-annotator agreement is left to future work.

Retrieval coverage. We additionally measure retrieval coverage as the fraction of the 38 classes for which the knowledge base returns at least one strongly relevant passage (cosine similarity ≥ 0.75) when queried by class name.

TABLE III
IN-DISTRIBUTION RESULTS ON HELD-OUT PLANTVILLAGE IMAGES (114 IMAGES, THREE PER CLASS ACROSS 38 CLASSES). ALL VALUES ARE PERCENTAGES; BEST IN EACH COLUMN IN BOLD. THESE NUMBERS ARE A CEILING REFERENCE AND LARGELY REFLECT MEMORIZATION OF THE TRAINING DISTRIBUTION (SEE TEXT).

Config	Acc	F1	Corr.	Harm.	Naive	Interv.
FedAvg	**100.0**	**100.0**	**99.1**	**0.0**	0.0	**0.9**
FedOpt	95.6	95.3	97.4	0.9	4.4	3.5
FedMA	86.0	82.8	87.7	3.5	14.0	17.5
Ensemble	**100.0**	**100.0**	97.4	**0.0**	0.0	2.6

VI. Results and Discussion

We evaluate on two test sets: an in-distribution set drawn from the same PlantVillage source used to train the backbones (Table III), and an out-of-distribution set of web-sourced images (Table IV). Both are reported for each individual model and for the three-model ensemble. Fig. 2 visualizes the accuracy gap between the two settings, and Fig. 3 the safety improvement of the agentic layer under distribution shift.

Generalization gap. On the in-distribution PlantVillage images (Table III) the models score very high – 86.0–100% top-1 accuracy for individual models and 100% for the ensemble. These same-source numbers, however, largely reflect memorization of the training distribution rather than field-relevant recognition, and we report them only as a ceiling.

TABLE IV
OUT-OF-DISTRIBUTION RESULTS ON WEB-SOURCED IMAGES (190 IMAGES, FIVE PER CLASS ACROSS 38 CLASSES). ALL VALUES ARE PERCENTAGES; BEST IN EACH COLUMN IN BOLD.

Config	Acc	F1	Corr.	Harm.	Naive	Interv.
FedAvg	20.5	17.4	72.1	24.2	79.5	41.1
FedOpt	23.2	**20.4**	70.5	22.6	76.8	44.7
FedMA	21.6	15.4	72.1	15.3	78.4	67.4
Ensemble	**24.2**	20.3	**74.7**	**14.2**	**75.8**	63.2

On the out-of-distribution web images (Table IV) the accuracy drops sharply, to 20.5–23.2% for individual models and 24.2% for the ensemble – roughly a four-to-five-fold gap, illustrated in Fig. 2. This drop is expected – the backbones were trained on PlantVillage's controlled, uniform-background images and are here confronted with heterogeneous web photographs – and it makes the out-of-distribution set a realistic stress case for autonomous deployment rather than a saturated benchmark.

Value of the agentic layer. It is precisely under this distribution shift that the safety contribution of the agentic layer is most visible. A naive policy that acts on every prediction would issue harmful (incorrect) treatment advice on 75.8–79.5% of cases. The agentic pipeline reduces this to 14.2–24.2% by recognizing low confidence, low margin, or model disagreement and deferring those cases to a human. For the ensemble, the harmful-action rate falls to 14.2% – a roughly five-fold reduction from its 75.8% naive baseline – at a human-intervention rate of 63.2%. This is the intended safety-coverage trade-off: on inputs where the perception module is unreliable, the system withholds automated advice rather than acting on a likely-wrong prediction.

Single model versus ensemble. The three-model ensemble dominates every individual model on the safety-relevant metrics, achieving the highest accuracy (24.2%) and correct-action rate (74.7%) and the lowest harmful-action rate (14.2%). Beyond accuracy, ensembling supplies an independent uncertainty signal: the three members disagreed on 143 of the 190 images, and this disagreement channel caught one confident error that confidence thresholding alone would have missed. Ensembling therefore improves both raw perception and the reliability of the deferral decision.

Grounding and coverage. Retrieval coverage is complete: all 38 classes return strongly relevant passages from the knowledge base. Qualitative inspection of the full-pipeline reasoning outputs found that they cite specific retrieved passages for their factual claims and, when the retrieved evidence described symptoms but lacked management guidance, the Reasoning Agent declined to fabricate a treatment and instead recommended expert consultation – the intended hallucination-averse behavior. Because the Orchestrator invokes the Retrieval and Reasoning Agents only for the subset of cases not resolved by the direct-lookup path, the full pipeline also incurs substantially lower average inference cost than a fixed design that always invokes retrieval and reasoning, though a full latency comparison is left for future work.

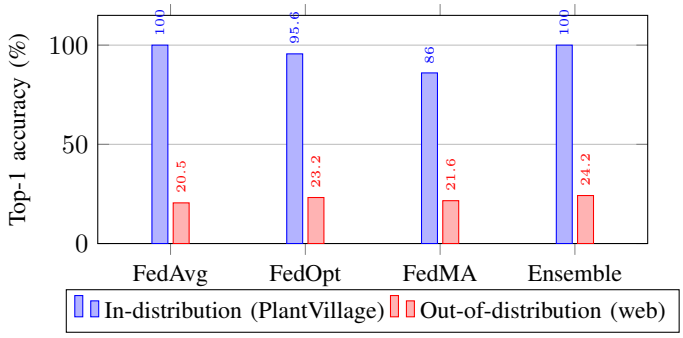


Fig. 2. Generalization gap: top-1 accuracy on in-distribution PlantVillage images versus out-of-distribution web images, per configuration. Every model collapses off-distribution, exposing the reliance of raw accuracy on the training distribution.

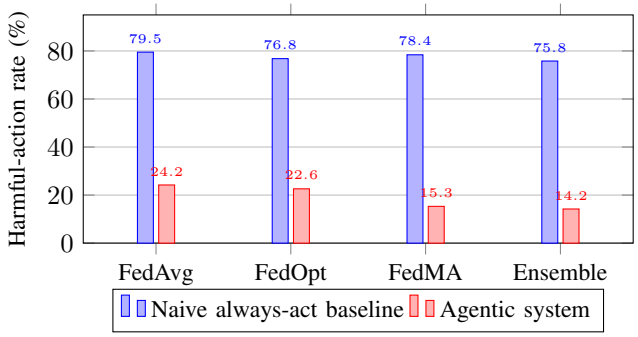


Fig. 3. Safety improvement on out-of-distribution data: the fraction of cases receiving harmful (incorrect) automated advice under a naive always-act policy versus the agentic system, which defers uncertain cases to a human. The agentic layer cuts harmful actions three- to five-fold across all configurations.

A. *Limitations*

Three limitations qualify these results. First, absolute classification accuracy on the out-of-distribution set is low (24.2% for the ensemble), reflecting the substantial domain gap between PlantVillage's controlled training images and heterogeneous web-sourced photographs with cluttered backgrounds and variable lighting. The Vision Agent was reused unmodified, without any domain adaptation or fine-tuning on field data; closing this perception gap – through domain-adaptive training or field-collected training data – is the most direct route to improving end-to-end utility. The agentic layer mitigates the safety consequences of this gap but cannot substitute for a more robust perception module. Second, the test set is modest in size (five images per class); a larger, curated field dataset with verified labels would tighten the estimates and enable per-class error analysis. Third, the Orchestrator's confidence, margin, and agreement thresholds were set conservatively by design rather than tuned against a validation objective; the high human-intervention rate (63.2%) is a direct consequence of this caution, and a learned or calibrated routing policy, noted in Section VII, may achieve a more favorable safety-coverage trade-off than the fixed thresholds used here.

VII. Conclusion and Future Work

This paper presented an agentic AI framework that upgrades a standard CNN-based plant disease classifier into an autonomous, explainable, and safety-aware diagnostic advisory system. By decomposing the diagnostic process into a Vision Agent, a Confidence and Image-Quality Agent, a Retrieval Agent, a Reasoning Agent, and an Orchestrator that conditions computation and action on predictive uncertainty, the system moves beyond single-shot classification toward the autonomy, goal-directed reasoning, and environment-aware deferral that characterize agentic AI. The introduced agentic metrics – correct-action rate, harmful-action rate, and human-intervention rate – offer a more complete picture of deployment safety than classification accuracy alone, and our results demonstrate a substantial reduction in harmful autonomous actions relative to a naive always-act baseline.

Several extensions follow naturally from the architecture proposed here. An episodic *Memory Agent* would allow the system to retain a history of diagnoses per field or farmer, enabling pattern detection across repeated visits. A *Follow-up Agent* could request a subsequent image after a fixed interval to track disease progression or treatment efficacy over time, an evaluation that PlantVillage's single-image structure does not currently support and would require new longitudinal data collection. Finally, the largest single gain available is not architectural but perceptual: closing the generalization gap exposed in Section VI through domain-adaptive training or field-collected imagery would raise end-to-end utility far more than any refinement of the agentic layer alone. Replacing the Orchestrator's fixed confidence, margin, and agreement thresholds with a learned or calibrated routing policy is a further promising direction toward deployable, trustworthy agricultural AI.

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